

URBAN SYSTEMS / CASE 07

VERTICAL CITY RETROFIT

**The next city is not built from scratch.
It is upgraded in place.**

A strategy for aging high-rise housing
envelope · energy · shared space · care

CONCEPTUAL HERO VISUAL / NOT A REAL PROJECT

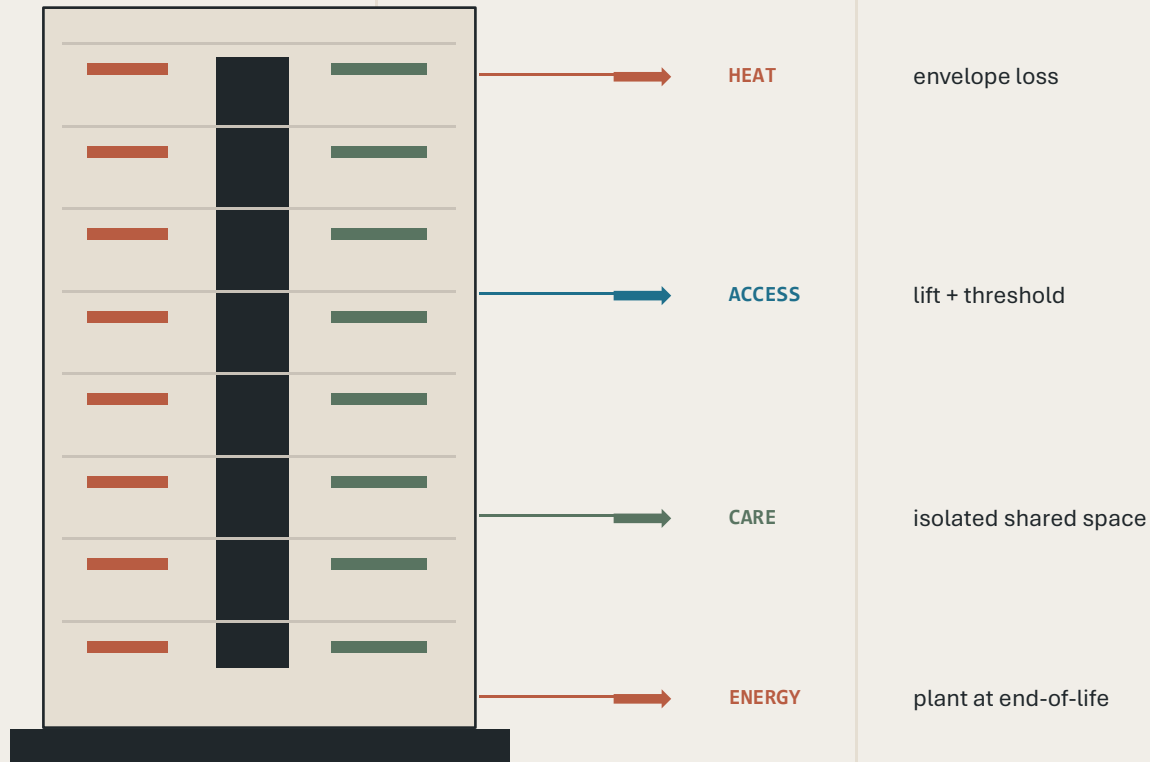
VERTICAL CITY RETROFIT · 2026



An aging tower is a stack of hidden pressures

The façade is only the visible layer; the retrofit brief begins where energy, access, and care intersect.

SECTIONAL DIAGNOSIS / EXISTING CONDITION

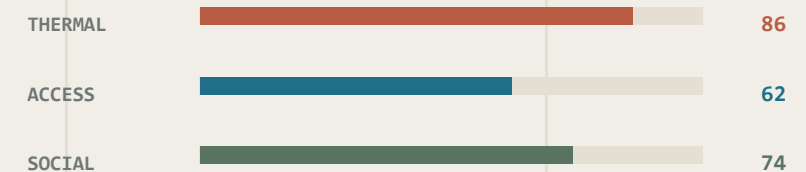


THE RETROFIT BRIEF

Upgrade the whole section.

A façade-only fix can lower heat loss, but it cannot repair the social and operational systems around the home.

PRESSURE INDEX / ILLUSTRATIVE



The unit of change is the building section.

One tower, five systems

The retrofit is not a façade package. It is a coordinated intervention across the whole vertical section.

EXPLODED AXONOMETRIC / OPERATING SECTION

01 ENVELOPE

thermal skin

02 ENERGY

plant + distribution

03 MOBILITY

lift + threshold

FROM BUILDING TO OPERATING MODEL

Value appears in the connections.

The same section can reduce loss,
restore access, and create care capacity.

04 SHARED SPACE

lobby + terrace

05 CARE

neighbourhood support

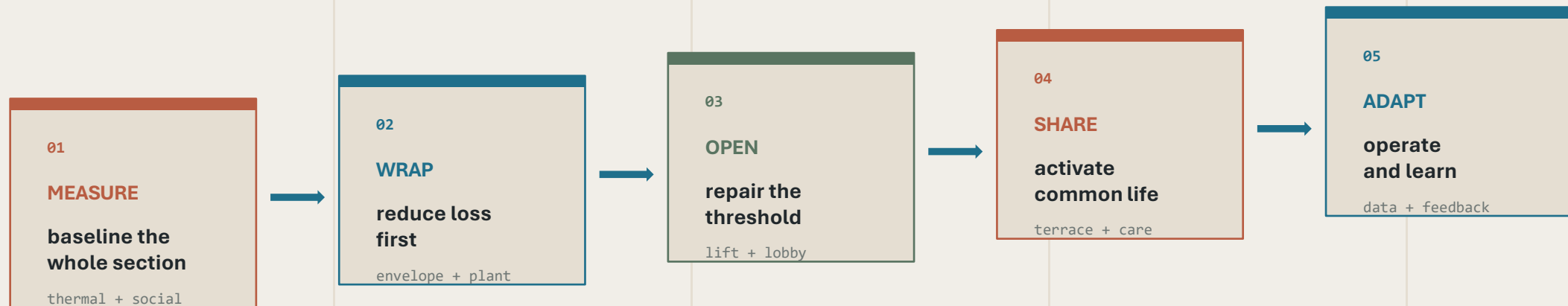
ONE SECTION / FIVE OPERATING
LAYERS

The tower behaves like a city when its layers
work together.

The intervention moves in five deliberate steps

A retrofit succeeds when each move unlocks the next one without interrupting the life of the building.

OCCUPIED-BUILDING SEQUENCE



THE DELIVERY TEST

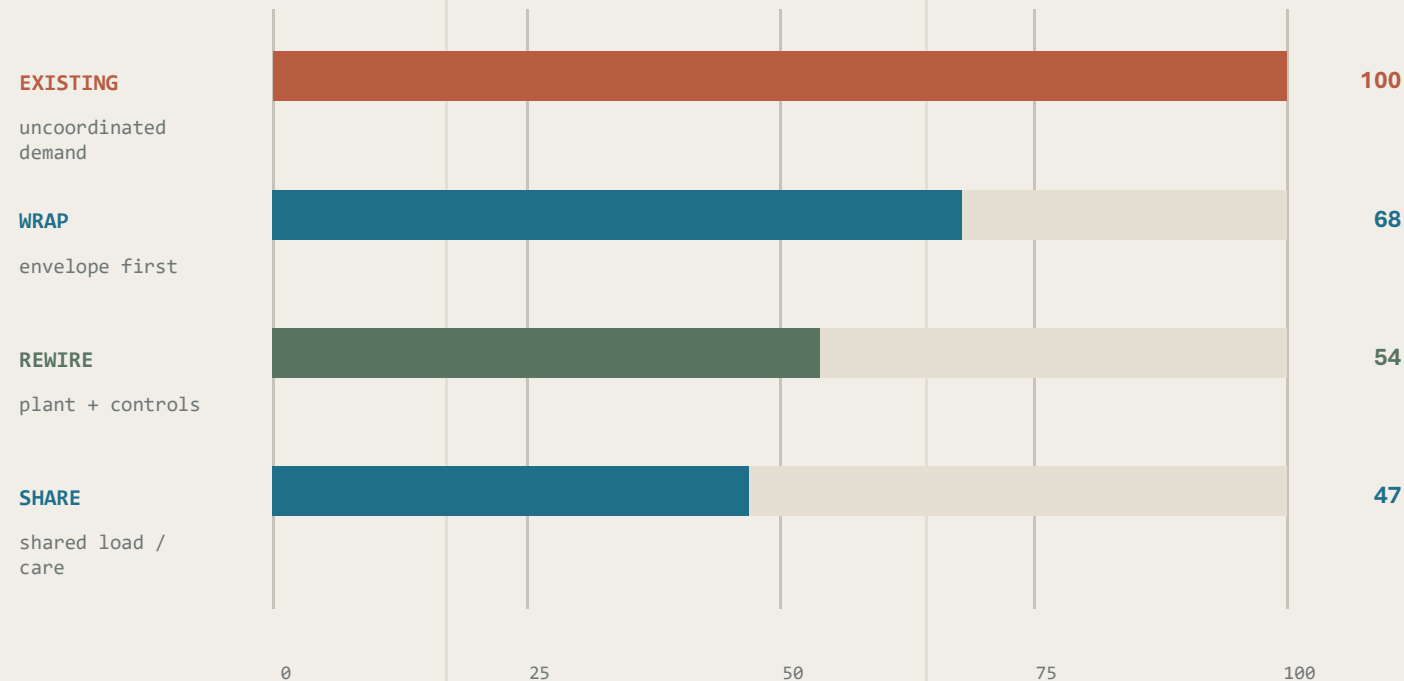
Can the building become more comfortable, more connected, and more useful while people remain inside it?

The sequence is the value proposition: reduce loss → restore access → create shared capacity → learn through operation.

Energy is the first constraint

Before adding new capacity, reduce the amount of energy the building needs to perform everyday life.

ILLUSTRATIVE LOAD INDEX / EXISTING = 100



Illustrative sequence: reduce loss → rewire capacity → share the gain.

THE STRATEGIC READ

Reduce loss before adding capacity.

A plant upgrade can improve efficiency; an envelope upgrade changes the demand itself.

01	DEMAND	lower the peak
02	CAPACITY	right-size the plant
03	RESILIENCE	keep options open

The first retrofit is a demand decision.

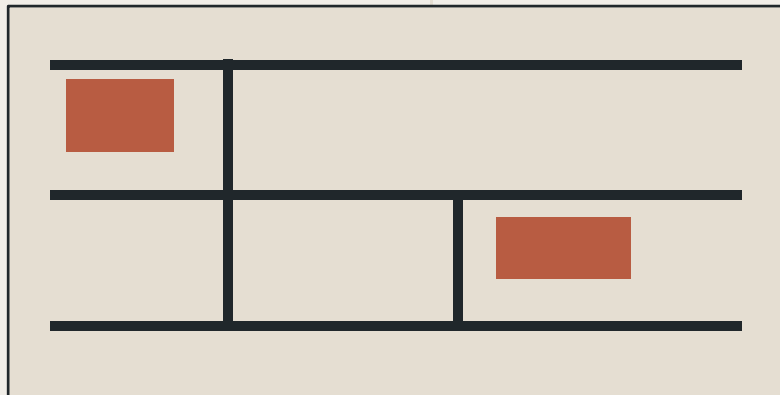
The ground floor is the social switch

A small change at the threshold can turn circulation space into shared civic capacity.

GROUND-FLOOR PLAN / BEFORE → AFTER

01 / EXISTING

sealed threshold

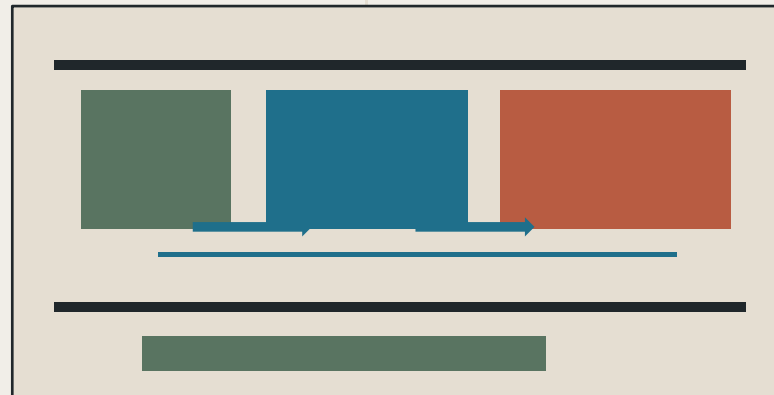


lobby / parking edge / no shared program

Movement passes through. Nothing stays.

02 / SHARED GROUND

porous threshold



lobby / care / terrace / flexible room

Movement meets. Something can happen.

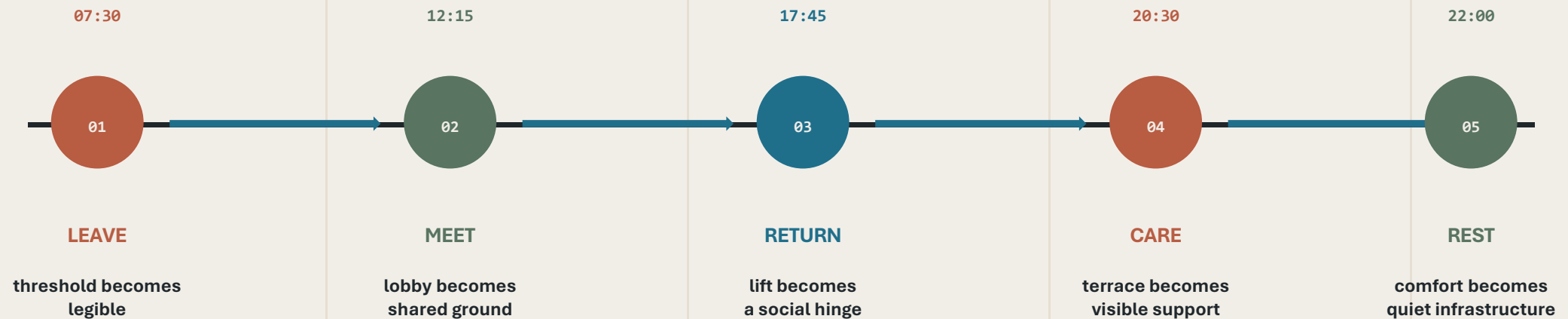
The retrofit does not only improve the envelope; it increases the number of useful encounters the building can support.

A vertical city is measured in moments

The retrofit succeeds when the building supports more useful moments across an ordinary day.

DAY STORYBOARD / OCCUPIED BUILDING

FROM MOVEMENT TO CAPACITY

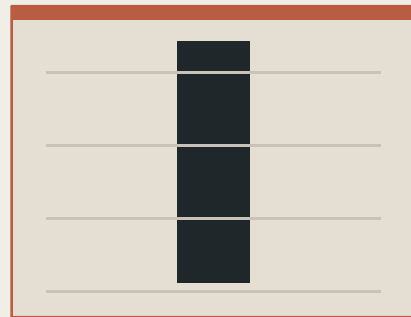


A retrofit is not finished when the construction ends; it is finished when daily life has more room to work.

There is no single retrofit

The right depth depends on capital, disruption, and the amount of shared capacity the building needs to recover.

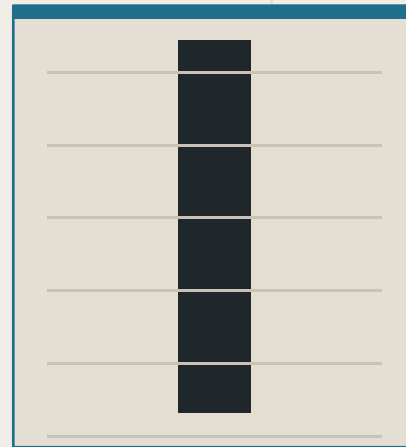
THREE INTERVENTION DEPTHS



LIGHT

protect

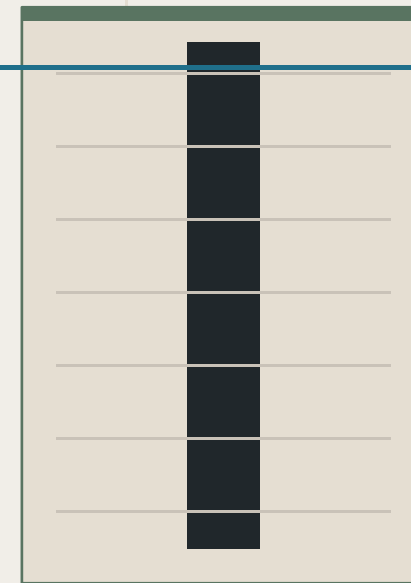
skin + controls



DEEP

reconnect

skin + threshold + care



DISTRICT

share

building + neighbourhood

SELECTION LOGIC

**Depth follows
shared need.**

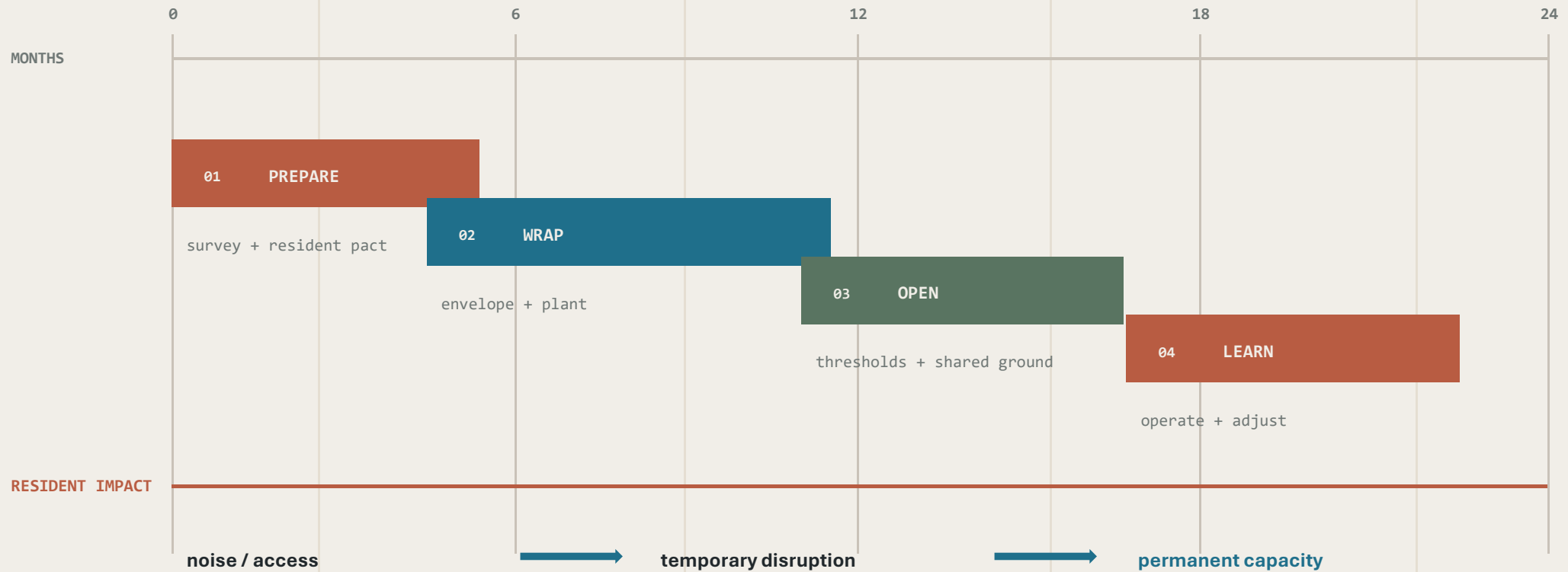
Light protects.
Deep reconnects.
District shares.

**Choose the smallest
intervention that changes the
operating model.**

The building stays occupied while the work moves

Construction phasing is part of the design: every intervention must have a resident-safe next move.

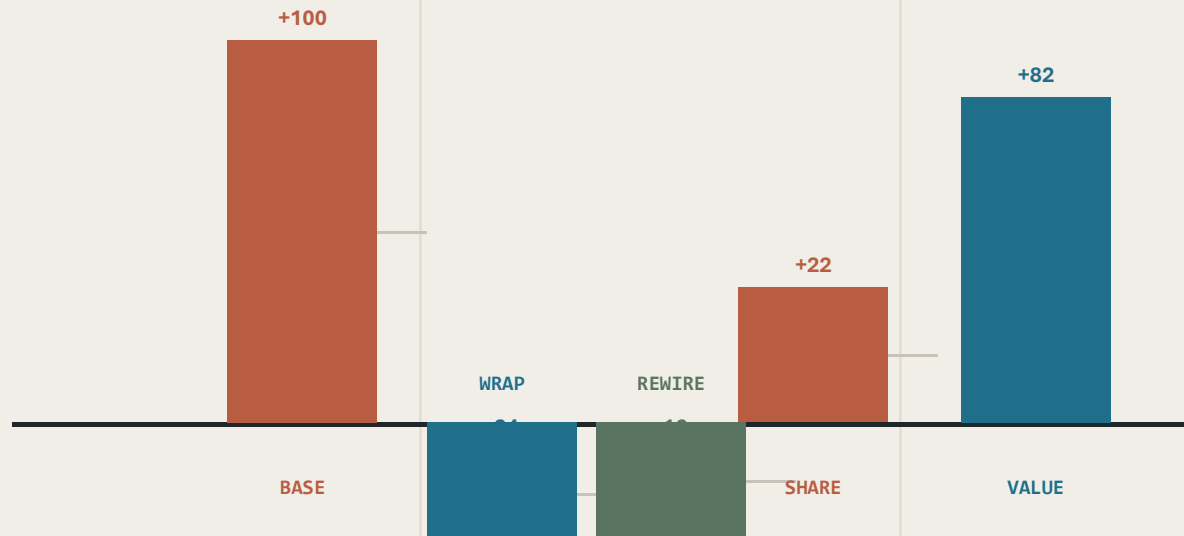
RESIDENT-IN-PLACE PHASING



Sequence changes the economics

The same interventions can produce different value depending on what happens first.

ILLUSTRATIVE VALUE BRIDGE / INDEX



Illustrative index: demand reduction creates the headroom for shared capacity.

THE ECONOMIC READ

Order is a financial lever.

Reduce peak demand.

Right-size capacity.

Invest in shared use.

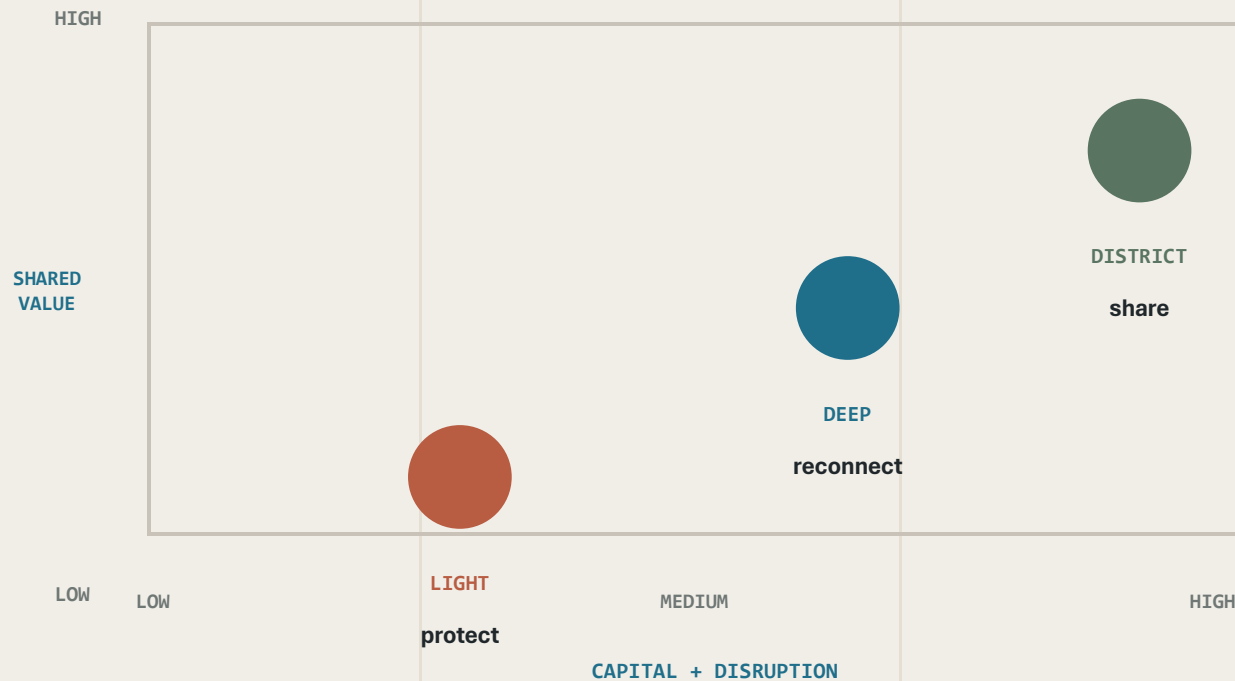
SENSITIVITY / WHAT MOVES FIRST

CAPEX	high
DISRUPTION	medium
SHARED VALUE	later

Choose by trade-off, not by score

A credible retrofit brief makes the conflicts visible before it recommends a depth.

OPTION MATRIX / STRATEGIC FIT



THE DECISION RULE

Pick the option that changes the system.

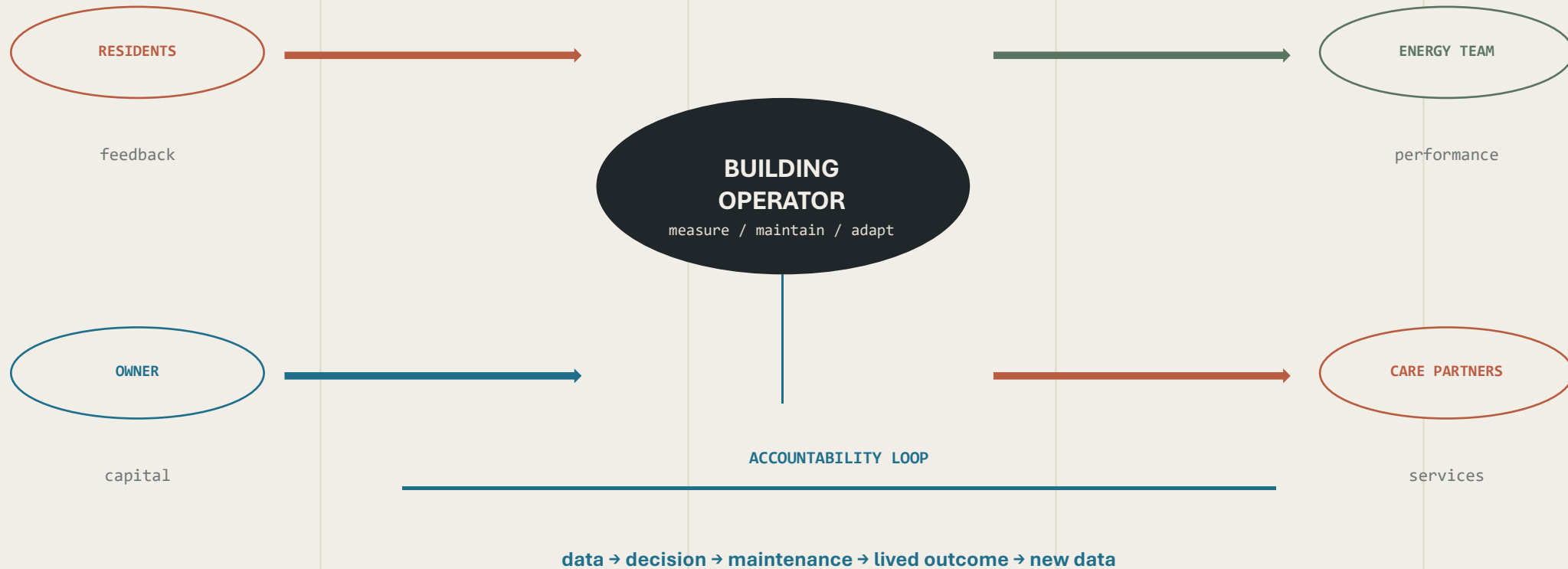
Light: minimum disruption
Deep: balanced value
District: maximum capacity

Trade-offs are not a weakness; they are the decision.

The retrofit needs an operator

Capital can upgrade the building once; an operating model keeps the new capacity useful.

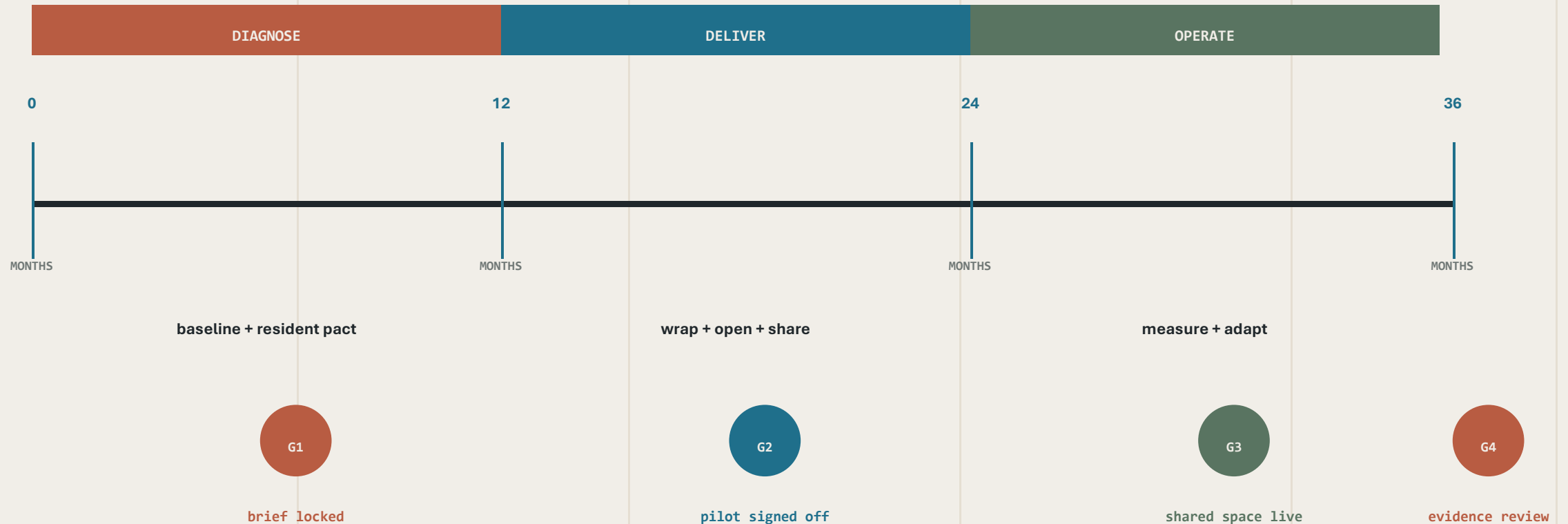
ACTORS / FLOWS / ACCOUNTABILITY



A 36-month path with evidence at every gate

The programme becomes investable when each horizon has a decision, an owner, and proof of progress.

THREE HORIZONS / FOUR GATES



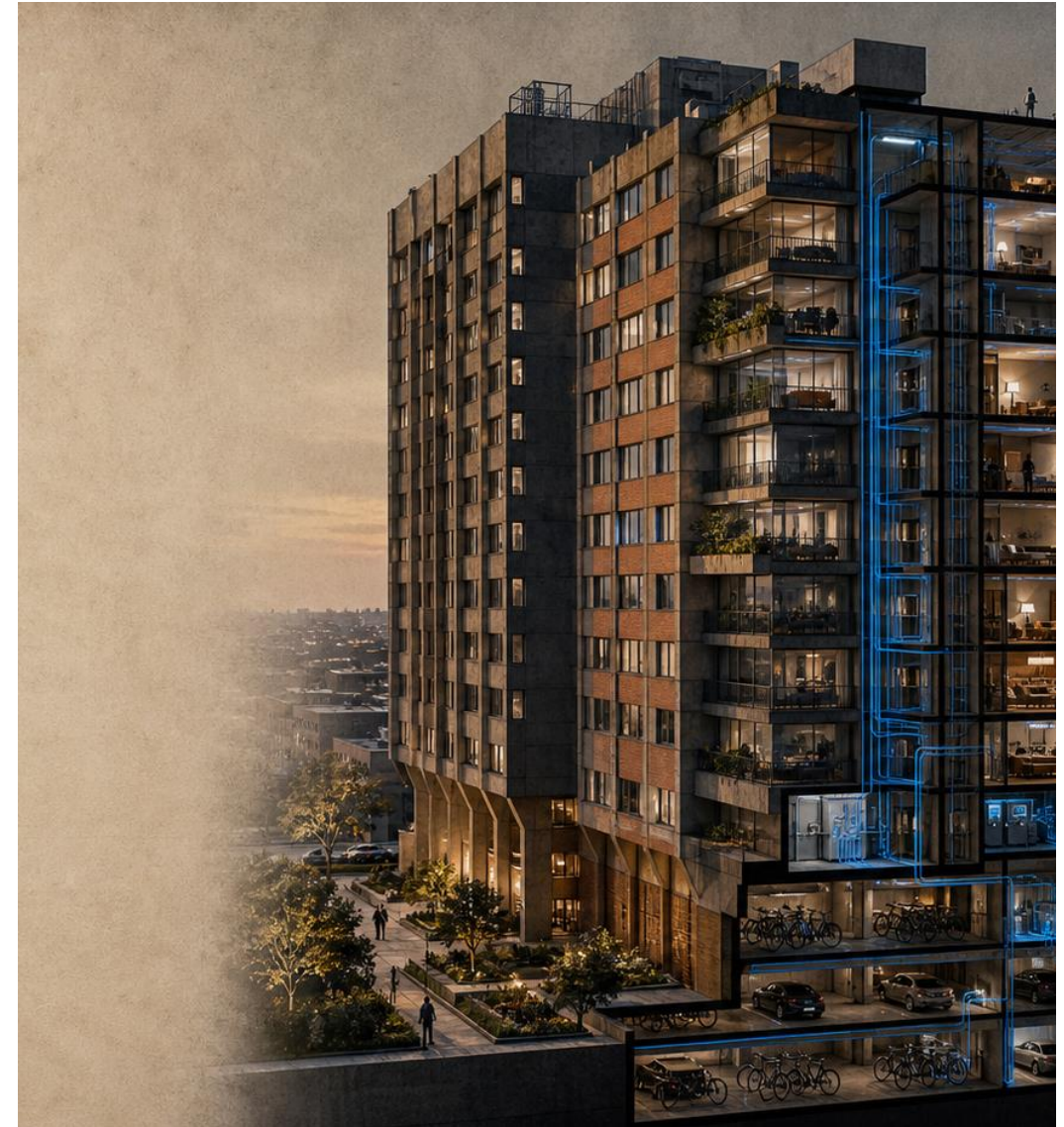
A gate is not a date. It is a decision supported by evidence.

Conceptual 36-month path; milestones and gates are illustrative programme design assumptions.

Upgrade in place. Keep life in place.

A retrofit becomes strategic when the building is treated as a connected vertical city—not a façade project.

- 01 **Reduce loss before adding capacity.**
- 02 **Design the shared ground.**
- 03 **Operate the new capacity.**



CONCEPTUAL HERO VISUAL / NOT A REAL PROJECT